The EIC and ePIC for a polarized-lithium programme: beam and detector parameters, their provenance, and the lithium case derived

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2026-08-27 · rewritten 2026-08-28 with the far-forward optics re-derived per configuration, restated 2026-08-29 on the per-configuration transport, corrected 2026-09-02 on the cross-report consistency review, re-sourced 2026-09-06 · the specification behind Reports 1, 2 and 4 · every entry carries its source and its **status** in the vocabulary of §1 · figures from `evgen/scripts/eic_beam_figures.py`, tables from `polli_fastsim.beams`, `polli_fastsim.farforward` and `fastsim/scripts/tagging_acceptance.py`

Abstract. Every projection of the polarized-lithium programme rests on parameters that belong to the accelerator or to the detector. This report collects them, states what each one is — a design value, a requirement, a measurement, a derivation of ours, or an unknown — and derives the lithium case where nothing is published. Three results carry the rest of the programme. Ion energies are set by the revolution period, not by rigidity: ions are γ-matched to the proton configurations, so the ^{6}Li menu is 41 and 99–138 GeV/u (precisely 40.8, and 99.5–137.5) with nothing between. The beam divergence, which every far-forward acceptance is an exponential in the square of, follows from the Yellow Report's proton tables with a species factor that is unity wherever the ion is γ-matched and rises only at the rigidity-capped top, to √2 for ^{6}Li: 220/380, 180/180 and 92/92 μrad (h/v) for ^{6}Li at the three configurations. Under those optics the intact-^{6}Li recoil cannot be tagged at any configuration (9×10^{-10} to 6×10^{-27} with the measured pot aperture) and the ^{6}Li α spectator only through its off-rigidity window and the over-rigid inner branch beside it (1.7–2.6%); a lithium tagging optics — the horizontal β* raised 46–164× with the pots following the envelope, at 1/6.8–1/12.8 of the luminosity — recovers a 25–37% coherent tag and a 22–31% α tag, and is the one *new* optics this programme asks for; ^{7}Li keeps the published one, for which it pays nothing, so the ask is two run configurations rather than two lattice designs. The detector model is a requirement where a requirement exists and a labelled stand-in where none does: the track angular resolution and the electron-identification curve have no published ePIC source.

1 · Scope and status vocabulary

Report 1 presents the measurement and Report 2 the simulation and reconstruction chain; this report is the specification both are conditional on, in a form that separates what the machine and detector documents state from what this programme has had to derive or assume. Five status words are used throughout: **design value**, a parameter that appears in a design document; **requirement**, a specification a subsystem must meet, not a demonstrated performance; **measured**, from test beam, operations, or — for the pot aperture — a simulation of the detector geometry; **derived**, computed here from published inputs with the derivation shown; and **unknown**, not published anywhere we could reach. Where no published value settles a parameter, the status names what this programme puts in its place instead — an **assumption** adopted for the projections, a **placeholder** or **stand-in** of the right magnitude, or this programme's own **proposal** — and Table 8, being a fact-check, uses the same column to say how each value fails its source. §2 gives the collider and the ion energy menu, §3 the beams, §4 the lithium case, §5 the far-forward acceptance of the lithium tags at each optics, §6 the ePIC detector, and §7 the list of what the programme assumes on the machine's and the detector's behalf, with owners.

2 · The collider and the ion energy menu

The EIC is two rings in the RHIC tunnel: the Electron Storage Ring at 5, 10 and 18 GeV and the Hadron Storage Ring carrying protons to 275 GeV and ions to the same magnetic rigidity, 917.3 T·m. ePIC sits at IP6; a second interaction region at IP8 is designed but not funded. The crossing angle is 25 mrad at IP6 and 35 mrad at IP8, recovered by crab cavities [1,2].

Colliding bunches requires equal revolution periods. The electrons are ultrarelativistic at every energy, so their period is fixed; the hadrons must match it, and since the circumference is adjustable only within a narrow range, the hadron velocity — hence γ — is quantised by the ring geometry, and the magnets supply whatever rigidity that γ implies for the species in the ring, up to the cap. Two limits therefore bind at the two ends of the menu: at the top the magnets bind and the ion is rigidity-capped at p/nucleon = 275 GeV × Z/A; at the bottom the circumference binds and the ion is γ-matched to the proton configuration, at a per-nucleon energy that rigidity scaling does not give. GeV/u throughout is the per-nucleon momentum `beams.Ion.momentum_per_nucleon_at` returns, which at these γ is the per-nucleon total energy to better than 0.03%. EPIOS describes the mechanism [2]: a radial shift of up to ±20 mm in the arcs covers 118 < γ < 293, and a bypass arc 90 cm shorter reaches γ = 43.5. The check that settles the rule is Yellow Report Table 10.2, which lists gold at 41 GeV/u — γ = 44.02 against the 41 GeV proton's 43.70, equal to 0.7%, the u-versus-m_p mass difference — where rigidity scaling would give 16.4 GeV/u; the same rule reproduces the published ^{3}He menu, "41 GeV/nucleon, and from 100 to 183 GeV/nucleon" [9], exactly. One conflict is recorded rather than resolved: Yellow Report Table 10.1 runs 100 GeV protons, γ = 106.6, in neither stated window. The programme stays anchored on the Yellow Report's three configurations because every beam-parameter table is indexed by them, and `tools/consistency_check.py` keeps the point visible.

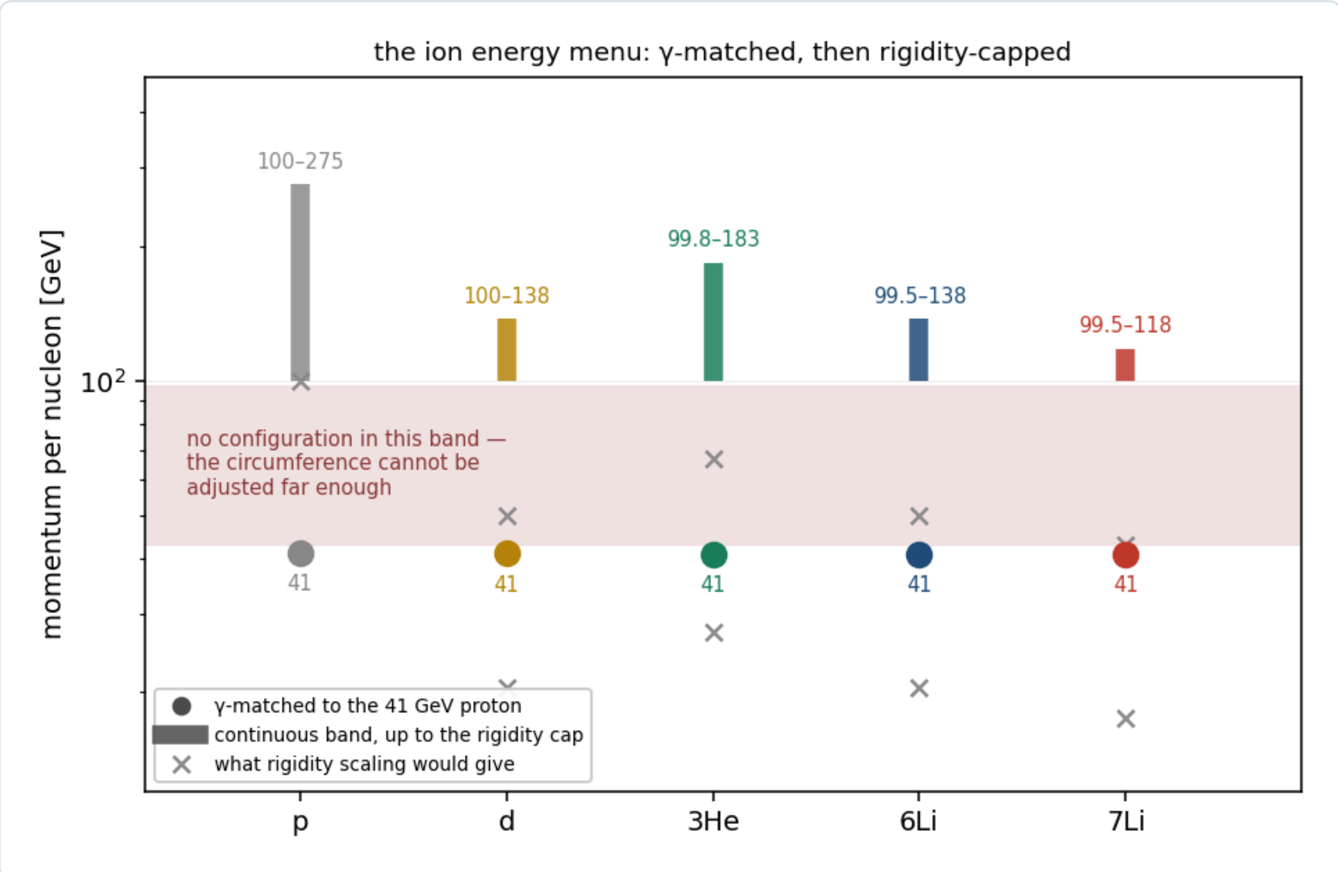

Figure 1 — the ion energy menu. Each of the five species plotted has an isolated γ-matched point at ≈ 41 GeV/u and a continuous band from ≈ 99 GeV/u to its own rigidity cap, with nothing between. Crosses mark what rigidity scaling would have given, which this programme used until 2026-08-27. `eic_beam_figures.py`.

SPECIES	A/Z	Γ-MATCHED POINT	CONTINUOUS BAND	RIGIDITY CAP	STATUS
p	1	41 GeV	100–275 GeV	275 GeV	design value
d	2	41.0 GeV/u	100–137.5	137.5	design value (YR Table 10.3 prints 135)
^{3}He	3/2	40.9	99.8–183.3	183.3	design value (menu published [9])
^{6}Li	2	**40.8**	**99.5–137.5**	137.5	**derived**; top ≈ 138 in EPIOS
^{7}Li	7/3	**40.8**	**99.5–117.9**	117.9	**derived**; top ≈ 117 in EPIOS
Au	2.49	41	– (110 published)	110.3	design value

Table 1 — the energy menu. `beams.default_configs` computes these; run on a proton it returns 41 / 100 / 275 exactly, the implementation's self-check. Only the lithium rows are derived, and only their top energies appear in EPIOS [2].

3 · The beams

3.1 · Electron beam

QUANTITY	VALUE	SOURCE	STATUS
energies	5, 10, 18 GeV	YR §10.1	design value
β* (h/v)	45 / 5.6 cm	EPIOS Table I (EIC column: 10 × 275 GeV, 1160 bunches)	design value
RMS emittance (h/v)	20 / 1.3 nm	EPIOS Table I	design value
bunch charge, bunches	28 nC, 1160	EPIOS Table I	design value
divergence h/v, 18 GeV	202 / 187 (HD), 89 / 82 μrad (HA)	YR Table 10.1	design value
Δp/p	$5.8–10.9 \times 10^{-4}$	YR Table 10.1	design value
polarization	70%	YR physics studies	assumption for projections; unused by the tensor observable

Table 2 — the electron beam. The EPIOS rows are the 10 × 275 GeV, 1160-bunch column and the divergence row is the 18 GeV one, so the two do not combine: $\sigma_\theta = \sqrt{\varepsilon/\beta^*}$ on the EPIOS pair returns 211 / 152 μrad, which is that column's own Yellow Report entry, not the 202 / 187 above. The electron divergence enters this programme only through the track angular resolution of Report 2 §3, for which it is the irreducible floor.

3.2 · Hadron beam

Yellow Report Tables 10.1 (e+p) and 10.2 (e+Au) are the authoritative divergence and momentum-spread tables and carry both optics settings; high divergence and high acceptance differ by a β* choice, and the Yellow Report states that $\sigma_\theta \propto 1/\sqrt{\beta^*}$ while the luminosity goes as 1/β*, so acceptance is bought with luminosity [1]. The baseline cooling is conventional electron cooling at injection, demonstrated at LEReC; EPIOS is explicit that store cooling is a later-stage addition worth a factor two in average luminosity [2], so the emittance grows under intrabeam scattering through the fill with nothing opposing it. The Yellow Report's proton table is captioned for strong hadron cooling, which would hold the emittance at its design value through the store; that design value, 11 / 1 nm, is the emittance EPIOS states injection cooling creates (at the design β* it reproduces Table 3's 275 GeV row, 117 / 118 against 119 / 119 μrad), so under the baseline the divergences of Table 3 — and every far-forward acceptance derived from them in §4 and §5 — are start-of-store values, the smallest a fill sees. The other design values (EPIOS Table I, EIC column): β* = 80 / 7.2 cm, RMS emittance 11 / 1 nm, bunch charge 11.0 nC, 1160 bunches, bunch length 6 cm, peak luminosity 10^{34} cm^{-2} s^{-1}. The beam is flat, $\varepsilon_x/\varepsilon_y \approx 11$, with β* matched so that the two divergences come out nearly equal — which is why Table 3's h and v agree at 275 and 100 GeV but not at 41.

PROTON CONFIGURATION	HD H/V [MRAD]	HA H/V	ΔP/P [10^{-4}]	L [10^{33} $CM^{-2}S^{-1}$] HD / HA
275 GeV (vs e 18), 290 bunches	150 / 150	65 / 65	6.8	1.54 / 0.32
275 GeV (vs e 10), 1160 bunches	119 / 119	65 / 65	6.8	10.0 / 3.14
100 GeV	220 / 220	180 / 180	9.7	4.48 / 3.14
100 GeV (vs e 5)	206 / 206	180 / 180	9.7	3.68 / 2.92
41 GeV	220 / 380	220 / 380	10.3	0.44 / 0.44

Table 3 — YR Table 10.1, hadron rows, verbatim (design values; the Yellow Report's strong-hadron-cooling scenario, which under the injection-cooling baseline of §3.2 is the start of the store). At 41 GeV the two optics are identical: the only configuration with no separate high-acceptance option, and the one the coherent programme would most like one at (§5). The two 275 GeV rows differ by the electron beam they collide with, so 18 × 137.5 inherits the 290-bunch row: every high-divergence number in this programme stands on 150 / 150 μrad there, ^{6}Li 212 / 212 by §4.1, while the high-acceptance 65 / 65 is common to both rows.

GOLD CONFIGURATION	STRONG HADRON COOLING H/V	ΔP/P	STOCHASTIC COOLING H/V	ΔP/P
110 GeV/u	218 / 379 μrad	6.2×10^{-4}	77 / 380	10×10^{-4}
41 GeV/u	275 / 377	10×10^{-4}	174 / 302	13×10^{-4}

Table 4 — YR Table 10.2, gold rows. The two cooling scenarios differ by 2.8× horizontally at 110 GeV/u: the cooling choice is worth as much as the optics choice.

4 · Lithium: published, derived, unknown

What is published is thin. EPIOS [2] gives the spin factors and the top energies and states the case; it gives no lithium luminosity, emittance or intensity, and describes the polarized-lithium source as early-stage. Hamwi and Hoffstaetter [3] give the depolarizing-resonance spectrum and the statement that small-G ions cannot use Siberian snakes.

QUANTITY	^{6}LI	^{7}LI	SOURCE / STATUS
spin, A/Z	1, 2	3/2, 7/3	nuclear data
G factor	−0.178 (EPIOS), −0.1818 (Hamwi)	+1.532 / +1.5196	published; the two disagree in the last digits
max |Gγ|, resonances to cross	27, 81	191, 573	Hamwi Table I, design study
snake scheme	^{6}Li deuteron-like — not amenable to Siberian snakes; partial solenoid and tune jumps; ^{7}Li partial snakes		Hamwi; EPIOS
polarization in the ring, P_z / P_{zz}	0.7 / 0.6	0.7 / –	placeholder for every projection (§7 item 10); the ANL source targets are $P_z \geq 0.90$ and $P_{zz} \geq 0.80$, and the transport to the ring is **unknown**
energy menu	41, and 99–138 GeV/u (40.8, 99.5–137.5; Table 1)	41, and 99–118 (40.8, 99.5–117.9)	derived (§2)
σ_θ h/v, high acceptance	220/380, 180/180, 92/92 μrad	220/380, 180/180, 99/99	derived (§4.1); start-of-store emittance (§3.2)
Δp/p	$7–10 \times 10^{-4}$	same	derived; species-insensitive (RF bucket)
tagging optics (§4.2)	β^*_x × 46 / 164 / 89; envelope 0.36 × 3.8, 0.19 × 1.8, 0.12 × 0.92 mrad; L/L_{HA} = 1/6.8, 1/12.8, 1/9.5	β^*_x × 62 / 220 / 102; envelope 0.32 × 3.8, 0.18 × 1.8, 0.12 × 0.99 mrad; L/L_{HA} = 1/7.9, 1/14.8, 1/10.1 – the ^{7}Li beam's own optimum, not a second request (§5)	this programme's proposal
emittance, intensity, β*, luminosity	– nothing published –		unknown (§7)

Table 5 — the lithium beam.

4.1 · The divergence, derived

$\sigma_\theta = \sqrt{\varepsilon_N/(\beta\gamma\,\beta^*)}$. At a given machine configuration β* is common to every species, the lattice being set by rigidity, so the only species dependence is through βγ; and a γ-matched ion has the proton's βγ exactly (§2). The species factor on the proton divergence is therefore

$$\frac{\sigma_\theta^{\rm ion}}{\sigma_\theta^{p}} = \sqrt{\frac{(\beta\gamma)_p}{(\beta\gamma)_{\rm ion}}} = 1 \;\;(\gamma\text{–matched}:\; 5\times41,\; 10\times100), \qquad \sqrt{2}\;\;(\text{rigidity capped},\; A/Z=2:\; 18\times275), \qquad 1.52\;\;(A/Z=7/3)$$

— not a blanket factor, which an earlier version of this programme's correction assumed. Whether equal ε_N is defensible for lithium is tested on gold, the published case: scaling the 275 GeV proton of Table 3's 290-bunch row to Au at 110

GeV/u by $1/\sqrt{(\beta\gamma)}$ alone predicts 236 µrad against the observed 218 (h) and 379 (v), so $\varepsilon_N(\mathrm{Au})/\varepsilon_N(\mathrm{p}) = 0.85$ horizontally and 2.6 vertically. Intrabeam scattering at fixed beam current goes as Z^3/A^2: 0.75 for ^{6}Li, 1 for a proton and 12.7 for gold — 17× lithium's — and RHIC deuterons showed no measurable IBS growth while gold grew 20–45% per hour. Lithium is a proton-class ion; equal ε_N is the right starting assumption, and it is an assumption. Δp/p is the easy one: protons and gold give 6.2–13 × 10^{-4} across every energy and cooling scheme, set by the RF bucket rather than the species.

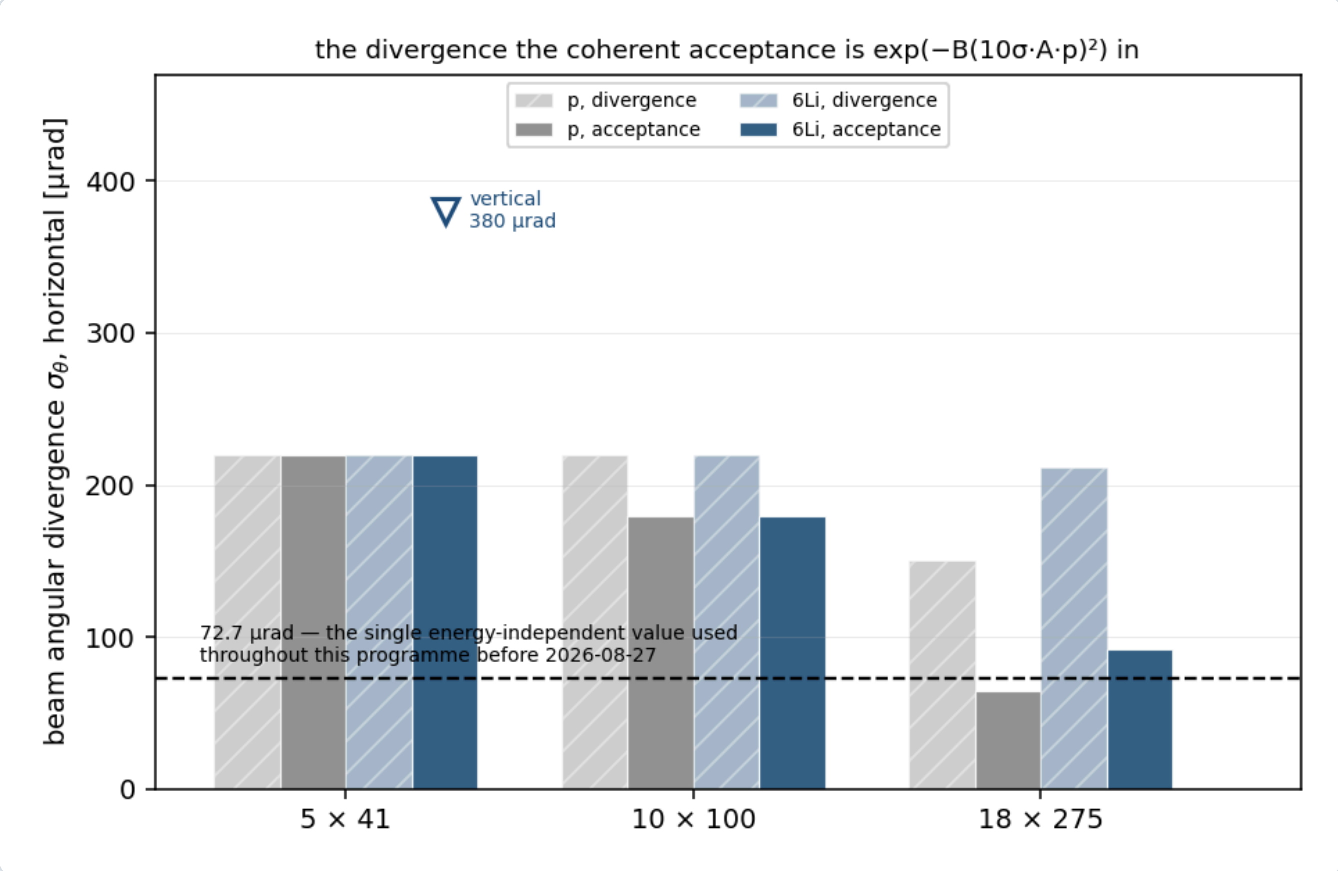

Figure 2 — the divergence. Protons and ^{6}Li per configuration and optics — the legend's "divergence" and "acceptance" are the high-divergence and high-acceptance settings of Table 3, and the 275 GeV pair is that table's 290-bunch row, 150 / 65 µrad, not the 1160-bunch row Report 1 §6.1 checks the β*–luminosity scaling against; the vertical value is marked where it differs from the horizontal. The species √2 appears only at 18 × 275; the dashed line is the single energy-independent 72.7 µrad on which every number published before 2026-08-27 rested (retired from the simulation on 2026-08-28), below almost everything. `eic_beam_figures.py` , `farforward.sigma_theta_for` .

4.2 · A lithium tagging optics

The far-forward lithium tags — the intact ^{6}Li recoil of the coherent channel and the α spectator of the tagged channels — are beam-blind (A/Z = 2 exactly; the α at 0.99813 of the beam rigidity) and are separated from the beam only by their angle at the interaction point against the 10σ envelope and the pot aperture. At the optics of Table 3 neither survives (§5). The one lever is β*: because a recoil escapes through the horizontal gap between the pots or through the vertical one, only the plane it escapes through needs a larger β*, and Report 1 §6.1 prices the horizontal de-squeeze alone — $\sigma_{\theta,x} \propto 1/\sqrt{r}$ at a luminosity $1/\sqrt{r}$, the vertical plane at high acceptance, the dispersion of the beam's momentum spread at the pots (D = 0.292 m against R_{12} = 29.97 m at 18 × 275, and 0.311 / 19.24 and 0.287 / 21.25 at the two lower configurations, all measured on 2026-08-28) added to the horizontal envelope, and the pots following the envelope in both planes. The optimum of tagged fraction × luminosity is at r = 46.5, 164.1 and 89.3 for the three ^{6}Li configurations, where the envelope is 0.36 × 3.8, 0.19 × 1.8 and 0.12 × 0.92 mrad, the tagged fraction 0.37, 0.25 and 0.33, and the luminosity 1/6.8, 1/12.8 and 1/9.5 of the high-acceptance value (`farforward.tagging_optics_point`). What that asks of the pots is a number: the horizontal envelope times the measured lever puts the inner edge at 6.99, 4.08 and 3.52 mm from the beam axis at the three configurations, against the 48, 32 and 16 mm at which the current geometry holds the horizontal inner edge of the silicon (the blind block of §5; in the vertical plane the envelope asks 17.3, 6.0 and 2.7 mm of pots that stand at the per-energy offsets of §6.1, 29.6, 7.1 and 2.7 mm, so that plane needs no move), so the ask is an approach a factor 6.9, 7.9 and 4.5

closer than today, the envelope widths of Report 4 §2, and it is the second half of the lever: the β* de-squeeze without it buys nothing, because the silicon and not the beam would then bind (Report 4 §2). Two assumptions carry it: the electron β* raised in step to keep the spots matched, and a parallel-to-point far-forward transport for the de-squeezed lattice ($R_{11}\sigma^* \ll R_{12}\sigma_\theta$ at the pots, as in the LHC's high-β* optics) — an assumption at the published optics as well, where the 10σ envelope is likewise read as $10R_{12}\sigma_\theta$ and R_{11} is unmeasured (Table 9 row 6), and the first thing to establish for a de-squeezed one. $\beta^*_x \approx 42$ m at 5 × 41 (46.5 × the 0.90 m horizontal β* of that configuration, where the high-divergence and high-acceptance optics coincide in Table 3, and which neither beam table prints — Yellow Report Table 10.1 carries no β* column and EPIOS Table I only the 275 GeV pair; the other two ask more, not less, ≈ 1.5 × 10² m at 10 × 100 at fixed normalised emittance and ≈ 2.3 × 10² m at 18 × 275, since r is referred to a high-acceptance β^*_x several times the design value — 11 nm / (65 μrad)² = 2.6 m there, against the 0.80 m high-divergence point of §3.2) is the smallest absolute ask, and a gentler de-squeeze than either the LHC has demonstrated for forward physics (β* = 90 m and 2500 m against a nominal 0.55 m). Unlike the isotropic ×14.5 optimum of plans/10 (β* = 13 m), which would have shrunk the beam in the final-focus quadrupoles ($\beta \approx \beta^* + L^2/\beta^* \approx 15$ m against 29 m today for L ≈ 5 m), a 42 m de-squeeze leaves the horizontal β at the first quadrupole at ≈ 43 m, larger than today, so the IR aperture joins matching and chromaticity among the questions for C-AD. It is a proposal, not a specification, and the request is stated in §7.

5 · The far-forward lithium tags at each optics

Table 6 evaluates both tags at the Yellow Report optics of each configuration and at the tagging optics of §4.2 with the pots following the envelope. The coherent tag is the analytic acceptance of an exp(−B|t|) recoil, B = 50 GeV⁻² (band 40–60, Report 1 §6.2, across which the coherent row of Table 6 moves by many orders of magnitude at the fixed aperture and by ±12% at the tagging optics), outside the rectangular cutout, with the pot aperture measured in the ePIC geometry (§6.1) as a second constraint per axis at the Yellow Report optics (cutouts 2.50 × 6.49, 1.80 × 2.12 and 0.92 × 0.92 mrad, the 6.49 being the 29.6 mm insertion over the measured 4.56 m vertical lever rather than an observed edge — the 5 × 41 vertical plane is shut, §6.1); the spectator tags are evaluated against the envelope alone, which binds horizontally at 10 × 99.5 and 18 × 137.5 and is itself exceeded by the measured 2.50 mrad silicon edge at 5 × 40.8, and are the cluster model of `polli_fastsim.spectator` (β = 0.30 GeV, the short-range parameter of the two-parameter cluster momentum density, whose 0.20–0.40 band is the model uncertainty [8]) routed through the far-forward windows of `polli_fastsim.farforward` — Roman Pots at 0.60 ≤ R ≤ 0.95, off-momentum detectors at 0.45 ≤ R < 0.65, B0 at 5.5–20 mrad, the near-beam band |R − 1| < 0.05 tagged only outside the envelope, and an over-rigid branch at R > 1.05 that reaches the Roman-Pot silicon on the inner side of the bend whenever the dispersive displacement clears the per-configuration blind block, 48 / 32 / 16 mm, and stays inside the last module at 144 mm — with the rectangular envelope applied to each fragment's azimuth.

TAG	OPTICS	5 × 40.8	10 × 99.5	18 × 137.5 (^{6}LI) / 18 × 117.9 (^{7}LI)
coherent ^{6}Li recoil, tagged fraction	YR high acceptance + measured aperture	9.4×10^{-10}	6.2×10^{-27}	7.1×10^{-14}
	tagging optics, pots follow (L/L_{HA})	0.37 (1/6.8)	0.25 (1/12.8)	0.33 (1/9.5)
^{6}Li α spectator (d struck), tagged	YR high acceptance	0.019	0.017	0.026
	YR high divergence	0.019	0.017	0.025
	tagging optics	0.31	0.22	0.29
^{6}Li d spectator (α struck), tagged	YR high acceptance	0.093	0.081	0.139
	tagging optics	0.54	0.46	0.57
^{7}Li α spectator (t struck), tagged	YR high acceptance	0.969†	0.968	0.975
	tagging optics	0.987	0.991	0.994

^{7}Li t spectator (α struck), tagged	YR high acceptance	0.778	0.919	0.939

Table 6 — the far-forward lithium tags per configuration and optics († the 5 × 40.8 ^{7}Li α entry is an angle-only routing; the measured per-band insertion leaves 57%, station 2 only — see below.) (`tagging_acceptance.py`, `nearbeam_aperture_scan.py`; the Yellow Report rows 2026-08-28, the tagging-optics rows re-run 2026-08-29 on the per-configuration transport). At the Yellow Report optics the ^{6}Li α tag is the 1.51, 1.51 and 1.54 points that fall below R = 0.95 into the Roman-Pot window, an over-rigid inner branch worth 0.13, 0.16 and 1.00 points at the three configurations, a near-beam tail that clears the envelope in 0.20 points at 5 × 40.8, 0.01 at 10 × 99.5 and 0.08 at 18 × 137.5, and 0.02 points into the B0 at 5 × 40.8 — 1.86, 1.68 and 2.61% in all, which is what the column rounds; the ^{6}Li d tag — the deuteron shares the beam's A/Z = 2, so it too sits at R ≈ 1 — is the 5.7–5.9% in the same window plus 1.5, 2.1 and 7.1 points over rigid, a near-beam tail of 1.8, 0.2 and 1.0, and a further 0.3 points into the B0 at 5 × 40.8 alone, with nothing in the off-momentum detectors. The ^{7}Li α at R = 0.856 lands mid-window regardless of optics at 10 × 99.5 and 18 × 117.9; at 5 × 40.8 the per-energy insertion holds the 32–48 mm silicon band off to |y| ≥ 18 mm, so the same α misses station 1 entirely and reaches silicon only at station 2 and in 57% of events — a module edge, which an angle-only routing cannot see (`tools/fullsim`). The triton (R = 1.29) is over-rigid, which since 2026-08-28 is a measured destination rather than a gap: `farforward.route_charged` carries an over-rigid branch (code 6, RP-inner) because a full simulation of the current ePIC geometry puts an R = 1.286 triton on the Roman-Pot silicon in 60 of 60 events at every configuration, at dx = +66 mm on the inner side of the bend, followed by a deposit in the zero-degree calorimeter in 80–98% of the same events (`tools/fullsim`, plans/09 B1 [8]). The ^{7}Li α + t double tag is therefore available at IP6, as an RP-inner track plus calorimeter energy, and the triton row above is almost all that branch (0.745 / 0.915 / 0.933) and not the low-momentum tail, the remaining 3.1 points at 5 × 40.8 being the B0: it is dispersive rather than angular, so the tagging optics moves it by under 0.3 points, to 0.780 / 0.922 / 0.941. What the double tag does not yet have is the α + t overlap with its m-dependence, one of the theory inputs collected in `docs/note_7li_theory_questions.md`. The third column is the top configuration of each isotope, 137.5 GeV/u for ^{6}Li and the 117.9 GeV/u rigidity cap for ^{7}Li. Before 2026-08-28 the fast simulation applied a single proton-derived 73 μrad at every configuration, and applied it azimuth-blind as a circle. The divergence alone, with the envelope kept rectangular as it is now, gives 2.8% for the ^{6}Li α at 18 × 137.5 and 16.8% at 5 × 40.8, against 2.6% and 1.9% here; the two choices together, as they were applied and before the over-rigid branch entered every column, read 1.9% and 20% against 1.6% and 1.7%, and 1.85% is what was published. Two readings of the ^{6}Li α number coexist in this repository and neither stands in for the other. The rows above are `polli_fastsim.spectator`, one partial wave per channel — L = 0 for α–d — while the tagged generator behind Report 4 §2.1 (`polligen.tagged`) carries the full S + D expansion of the same channel: the S-wave radial is identical in the two, ⟨k⟩ = 0.1071 GeV/c, and the D wave is hard at ⟨k⟩ = 0.2778, so at P_D = 0.0867 the channel mean moves to 0.1219 and the off-rigidity slice from 1.5% to 2.5%. The D wave is not an optional tail but the tensor observable itself — set P_D = 0 and the α–d density is m-independent and A_{zz}^{tag} vanishes identically — so the tagged asymmetries are quoted on the S + D spectrum and this table on the S-wave one: the Roman-Pot slice of the rows above is 1.5% at every configuration, against 2.5–2.8% there. The two ^{7}Li densities are the same pure P wave, and the 0.7 points by which this table's 0.9690 at 5 × 40.8 exceeds the tagged script's 0.9617 is a difference of *acceptance definition*: the column above is 1 − lost, i.e. any far-forward system, while `tagged_polarimetry_7li.py` masks on the Roman Pots alone, and at 5 × 40.8 the B0 carries 1.1% of the ^{7}Li α and the over-rigid inner branch a further 0.2% — 1.2 points in all, of which the k ≤ 1.2 GeV/c on which the tagged model's momentum grid ends gives back 0.6. Appendix A carries the like-for-like comparison.

Two conclusions. The tagging inversion between isotopes at IP6 is optics-independent: the ^{7}Li α works at 97% at every optics — in angle; at 5 × 40.8 the measured per-band insertion leaves 57%, station 2 only, and the 0.969 of Table 6 is an angle-only routing there (caption) — while the ^{6}Li α does not work at any published one. And the tagging optics converts the ^{6}Li tags from beam-blind to usable — 25–37% of coherent recoils, 22–31% of α spectators, 46–57% of deuteron spectators — at a luminosity cost of 6.8–12.8, which is the trade Report 1 §6.1 prices and Report 4 examines from the detector side. One caution on reading the two ^{6}Li spectator rows: they are not independent channels but the two fragments of the same breakup, and sampled jointly (Report 4 §4.2) both are recorded in 17–25% of breakups at the tagging optics (0.25, 0.17 and 0.22 at 5 × 40.8, 10 × 99.5 and 18 × 137.5) — so four fifths of the α row there lies inside the deuteron row, while about half the deuteron row is deuteron-only, the deuteron being the systematically wider fragment; at the Yellow Report optics, where the joint fraction is below 0.03%, the two rows are effectively disjoint.

The luminosity cost is not shared evenly between the isotopes, and for ^{7}Li it is not paid for at all. The ^{7}Li α row moves from 0.968–0.975 to 0.987–0.994 under the tagging optics — ×1.02 — for a luminosity of 1/7.9, 1/14.8 and 1/10.1, so the de-squeeze is a factor 7.7–14.5 *net loss* in tagged ^{7}Li events and multiplies every ^{7}Li error bar by 2.78, 3.81 and 3.15 at equal running time (Report 4 §2.1). Those are the price of the ^{7}Li beam's own optimum of the same figure of merit, β^*_x × 62 / 220 / 102; in the ^{6}Li tagging optics of Table 5, the one new optics the programme asks for, ^{7}Li pays 1/6.8, 1/12.8 and 1/9.5 instead and every bar grows by 2.6, 3.5 and 3.0. The α tag that lands mid-window is the one the machine cannot improve. So the inversion in the first conclusion runs in both directions: ^{6}Li needs the de-squeeze and ^{7}Li must not have it, which makes the two isotopes separate runs at separate optics settings — the ^{6}Li de-squeeze and the published one — rather than one fill plan, a scheduling consequence to put to C-AD alongside the lattice questions of §7.

6 · The ePIC detector

6.1 · Far forward

SUBSYSTEM	POSITION	ACCEPTANCE	TECHNOLOGY	STATUS
B0	z ≈ 5.4–6.4 m	5.5 < θ < 20 mrad, any charged; γ	AC-LGAD + EMCal	design value
off-momentum detectors	z = 25.5, 27.0 m	θ < 5 mrad, 0.45 ≤ R < 0.65	AC-LGAD	design value (windows [6]); positions from the geometry file
Roman Pots	z = 32.55, 34.25 m	θ < 5 mrad, 0.60 ≤ R ≤ 0.95; inner edge optics-dependent	AC-LGAD, in vacuum	design value (windows [6]); positions from the geometry file (current main)
ZDC	z ≈ 37.5 m	θ < 4 mrad, neutrals	W/Si + crystal	design value

Table 7 — the far-forward systems. The pot and off-momentum positions are read from `compact/far_forward/roman_pots_eRD24_design.xml` and `offM_tracker.xml` on the current main branch [4] and are not the 26 / 28 and 22.5 / 24.5 m the Yellow Report, the ePIC wiki and several papers still quote; only the angular windows enter the routing.

The pot aperture, measured. An intact ^{6}Li shot through the ePIC far-forward geometry (`tools/fullsim` , `epic-main` at git 9aaa2969, 216 map points in p_T × azimuth plus four 0.05 mrad angular ladders of 117 points each — 468 ladder events per configuration, one event per point, no beam envelope) clears the pots at $|\theta_x| \gtrsim 2.50 / 1.51 / 0.53$ mrad in the 5 × 41 / 10 × 100 / 18 × 275 optics against $|\theta_y| \gtrsim$ (no vertical acceptance) / 2.12 / 0.92 mrad: the aperture opens horizontally, because the beamline images an IP angle onto the pot plane with a horizontal lever R_{12} = 19.24 / 21.25 / 29.97 m against a vertical R_{34} of 4.56 / 3.35 / 2.93 m — a factor 4.2, 6.3 and 10.2 — the 5 × 41 entry measured on 2026-08-29 through a scratch geometry with the four Roman-pot section offsets zeroed, the real insertion there shutting the vertical plane. The edges quoted are each band threshold — 48 / 32 / 16 mm horizontally, the per-energy insertion vertically — over its own measured lever, and the first accepted angle of the ladder reproduces them to 1–4%: 0.55 mrad against the 0.53 the 16 mm block gives at 18 × 275, on a ladder whose step is 0.05 mrad. Twenty-nine point six millimetres of insertion over that 4.56 m lever is the 6.49 mrad the code carries as the 5 × 41 vertical half-width, past the 5 mrad at which the far-forward routing ends, and it is the one entry of the set with no first-accepted-angle row to close on — the ladder stops at 6.00 mrad, short of the insertion — so it is an extrapolation of a measured lever and not a measured edge. The 16 × 16 mm modules and per-energy 10σ offsets of the current geometry are what these numbers are measured in; against the September-2024 reading the aperture halved at 18 × 275 and widened at 5 × 41, which reverses the ordering of §5: the silicon binds at 5 × 41 and the machine at 18 × 275, where the earlier geometry said the opposite. The 5 × 41 row carries one further condition, and it is the largest uncertainty in this paragraph: it stands on the ePIC baseline compact files, whose 5 × 41 lattice is the separately re-matched proton one, taken here — as an educated guess stated in §7 — to be the lattice a ^{6}Li fill at 81.6 GV would run in at twice the field, while the Z/A = 0.5 light-ion lattice carried as the alternative gives R_{12} = 29.81 m and a 1.61 mrad edge — 48 mm / 29.81 m, the threshold the code applies, the first hit landing at 1.60 mrad on +x and 1.70 on −x — a factor 1.55 in the lever and 0.64 in the aperture (`farforward.POT_LEVERS_LIGHT_ION_LATTICE` ; §7). The same ladders fix the pot's outer edge, which the acceptance tables do not: the last angle at which a debris-free primary track is reconstructed is 2.85 / 3.85 / 4.00 mrad, not the θ < 5 mrad of Table 7 (96 / 106 / 150 mm at these levers) nor the 144 mm / R_{12} = 7.5 / 6.8 / 4.8 mrad the module arithmetic gives — beyond it the ion has struck the pipe and the pots see debris (`farforward.THETA_RP_OUTER_MEASURED`). The routing keeps 5 mrad as its default so that no published acceptance moves under the measurement; Report 4 §4.2 prices the partner-fragment veto at the measured edge, where it costs five points at 5 × 41 and nothing at the other two. Status: **measured, current geometry (2026-08-28)**.

Sensor and readout. AC-LGAD at 500 × 500 μm pitch with a 100 × 100 μm metal electrode over a 30 μm active thickness, bump-bonded to EICROC; timing ≈ 35 ps per plane is a requirement, HPK AC-LGADs measure 20–35 ps in test beam; spatial resolution ≈ 140 µm from the pitch alone, ≈ 20 µm shown with charge sharing, ePIC's stated pot baseline binary 500 µm without it; EICROC provides a 10-bit 25 ps time-of-arrival TDC and an 8-bit 40 MHz SAR ADC for charge [5]. The insertion is quantised by 16 × 16 mm modules in arrays three modules tall, each array placed by its centre at 2.4 cm — its own half-height — plus a per-energy offset, so the inner edge of the silicon sits at the offset alone (0.27 / 0.71 / 2.96 cm at 18 × 275 / 10 × 100 / 5 × 41, the 2.7 / 7.1 / 29.6 mm the vertical envelope of §4.2 is priced against), with the geometry file's own comment that these "are the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have" [4]. Rates: a few Hz per channel in normal running, 30–50 kHz at ≈ 10σ from beam halo on STAR experience [7]; in-vacuum cooling ≈ 100 W per sensor layer; no published radiation fluence or dose.

6.2 · Central detector

QUANTITY	VALUE USED	SOURCE	STATUS
EMCal σ_E/E, backward ($\eta < -2$)	2%/√E ⊕ 1%	YR Table 8.20, 2%/√E ⊕ (1–3)% backward (Fig. 8.3 gives the 2%/√E and no constant); the $PbWO_4$ crystal expectation	requirement, the optimistic corner of the YR band — YR Table 3.1 asks 1%/E ⊕ 2.5%/√E ⊕ 1% over $-3.5 < \eta < -2.0$
... backward transition ($-2 < \eta < -1$)	7%/√E ⊕ 1%	YR Fig. 8.3	stochastic exact; constant 2× optimistic against YR Table 3.1
... barrel ($\|\eta\| < 1$)	10%/√E ⊕ 1%	YR Table 8.20 band (10–12)%/√E ⊕ (1–3)%	optimistic corner; constant 2–3× optimistic against the ePIC BIC requirement
... forward ($\eta > 1$)	7%/√E ⊕ 1%	—	**mis-sourced**: 7%/√E belongs to $-2 < \eta < -1$; latent, no sweet-spot electron is forward
... noise term	absent	YR Table 3.1 has 1–2%/E	missing; 2% at 1 GeV
tracking σ_p/p	$\sqrt{(a\,p)^2 + b^2}$, η-piecewise	YR Table 11.2 (p. 437); YR Fig. 8.3 and Table 8.20 give the same barrel and forward rows but ⊕ 2.0% and ⊕ 1.0% backward where Table 11.2 gives ⊕ 0.5%, and ePIC's own requirement table carries Fig. 8.3's constants [7]; ATHENA Table 4 quotes Table 11.2 as "Requirements"	requirement — barrel and $1 < \eta < 2.5$ exact in all three, the $-2.5 < \eta < -1$ constant Fig. 8.3's 1.0% and so 2× Table 11.2's, endcap constants padded; the ePIC full simulation gives 0.38% at 1 GeV/c in $-1 < \eta < 1$, 1.08% in $-2.5 < \eta < -1$ and 3.0% in $-3.5 < \eta < -2.5$ (single pions, five η ranges, Figure 8.9 of [7]) against this table's 0.50, 1.00 and 3.00, so the barrel smearing is ≈ 30% pessimistic while the two backward slices, which hold eleven of the twelve sweet spots, are right to ≈ 8%; the electron measurement of a Craterlake 23.12.0 sample gives 0.45% in $0 \le \eta \le 0.5$ [10]
tracking σ_θ	5 / 3 / 2 / 1 / 2 / 3 / 5 mrad by η	—	**placeholder** — no angular requirement in either YR table; also the model's only stand-in for the 81–211 µrad beam-divergence floor on θ′, so the honest bracket is a factor ≲ 3 (Report 2 §3)
electron identification	0.70–0.95 by η	constructed between ATHENA JINST 17 P10019 Table 5 and ECCE NIM A 1055 168464 §3.5.2	anchored at one point; the sources differ by ≈ 25 points in the barrel; cancels in the spin-state ratio
hadron-side coverage, thresholds, efficiencies, noise	tracks $\|\eta\| \le 3.5$, $p_T > 0.2$ GeV, 95%; calorimeters $\|\eta\| \le 3.7$; 50 MeV noise	—	stand-ins of the right magnitude (Report 2 Table 2); the ePIC nominal forward reach is 4.0
hadron-side resolutions	YR tables per region, 2% EMCal constant	YR requirement tables	requirement

Table 8 — the central-detector model, fact-checked against the Yellow Report. The four EMCal σ_E/E rows are the model's `EMCAL_YR_TABLE`: the default on the hadron side, where it runs with the Yellow Report's 2% constant (the table's last row), and a switch on the electron side, where the published inclusive numbers apply the backward-endcap row at every η (Report 2 Table 2). Every sweet spot of the programme puts the scattered electron in the electron-going hemisphere — η =

−3.4 to −0.9 over the twelve spots of the three configurations, of which −2.9 to −1.6 is the mid configuration's four — so the mis-sourced forward row is latent. Nine of the twelve sit in the backward endcap, where the table is exact; two, at $\eta = -1.72$ and -1.65, sit in the backward transition, where it is right to within 20%; and the twelfth — the (0.141, 14.3 GeV²) spot at 5 × 40.8, $\eta = -0.91$, the bin §5.2 of Report 1 measures best at that configuration — reaches the barrel, where the table is optimistic by 15–26% depending on which corner of the Yellow Report band is taken and where the tracker, at 0.6% against 4.5%, is the better measurement. The angular resolution is the entry to watch: the 0.1 mrad quoted in some ePIC talks is a smearing input, not a measured resolution, and would delete the beam-divergence floor the placeholder stands in for.

7 · What the programme assumes that the specification does not give it

#	WHAT IS MISSING	WHAT WE DO INSTEAD	OWNER
1	lithium beam divergence and $\Delta p/p$, per configuration and cooling scenario	derive from the proton tables with the γ-matched species step (§4.1) at the start-of-store emittance of §3.2; the growth through a fill without store cooling is not modelled	C-AD / ePIC FF WG
2	a lithium tagging optics: can IR6 be matched with the horizontal hadron β^* raised 46–164× — the electron β^* co-de-squeezed so ξ_p stays ≤ 0.015, the Conceptual Design Report's hadron beam-beam design limit [11] — and what are the IP-to-pot phase advances (parallel-to-point transport)?	assume the optimum of §4.2 exists; without it no ^{6}Li far-forward tag exists at IP6 (§5)	C-AD
3	lithium emittance, intensity, β^* and luminosity — nothing published	assume proton-class ε_N (gold calibration, Z^3/A^2 IBS scaling) and the Yellow Report per-nucleon luminosity (the flat 10 fb^{-1}/u-per-year placeholder of Report 1 §3.1, not the per-configuration values of Table 3)	C-AD / ANL source group
4	what physically sets the 10σ retraction — collimation hierarchy, halo lifetime, or convention — and pots that follow the envelope of a de-squeezed optics	assume 10σ and pots that move (§4.2, Report 4 §3)	C-AD / ePIC FF WG
5	EICROC input charge dynamic range in fC, and LGAD gain suppression at ≈ 9 MIP	assume the charge is usable; it sets the margin of the far-forward Z-identification rather than its existence — one bit per plane is within a factor 1.4 of the 8-bit optimum, and the α + d second hit vetoes 84% of the α fakes at the tagging optics with no dE/dx at all (Report 4 §4)	OMEGA-IJCLab / BNL AC-LGAD group
6	per-optics transfer matrix at the pot planes — R_{11}, R_{12}, R_{21}, R_{22}, D, D′, and which 5 × 41 lattice a ^{6}Li fill runs in	use the measured (R_{12}, R_{34}, D) = 19.24 / 4.56 / 0.311, 21.25 / 3.35 / 0.287 and 29.97 / 2.93 / 0.292 m; the 5 × 41 R_{34} was obtained on 2026-08-29 through a scratch geometry whose four Roman-pot section offsets are zeroed, the real insertion there shutting the vertical plane, and R_{11}, R_{21}, R_{22} and D′ are still unmeasured. The 5 × 41 triple is the ePIC baseline (proton) lattice, which we adopt for the ^{6}Li fill as an educated guess — the same magnets and orbit at twice the field for the Z/A = ½ beam — with the Z/A = 0.5 file at the same ring setting (29.81 m, ×1.55 in the lever and ×0.64 in the aperture) carried as the labelled alternative. What is asked here is confirmation of that choice, not a decision the published numbers wait on	ePIC FF WG
7	ePIC tracking angular resolution and electron identification — no published curves	η-piecewise stand-ins, flagged in the code and in Table 8	ePIC
8	the calorimeter noise and threshold floor on the hadronic $E - p_z$ sum at Σ_h = 0.2–0.5 GeV	50 MeV Gaussian stand-in; it sets the $y \approx 0.01$ bins (Report 2 §4.2)	ePIC inclusive WG
9	radiation environment at the Roman Pots — no published fluence or dose	not modelled	ePIC FF WG

10	polarization-preservation scheme for ^{6}Li, which cannot use Siberian snakes, and the spin direction it delivers at IP6: the stable direction of the arcs is vertical, but no rotator or snake configuration is defined for a lithium fill, and every cos 2φ projection of Reports 1 and 2 takes the alignment axis transverse, θ_S = 90°	assume a scheme exists and that the ring holds the projections' placeholders P_z = 0.7, P_{zz} = 0.6 (tensor fills +0.6 / −1.2), already derated from the EPIOS/ANL source targets $P_z \geq 0.90$, $P_{zz} \geq 0.80$; every tensor error scales as $1/P_{zz}$; and assume the vertical axis of the arcs is the axis delivered at IP6 (plans/04 #2), a polar tilt diluting the amplitude by $\sin^2\theta_S$ — 0.8% at 5° (Report 2 Table 2) — and no transverse axis leaving no observable at all	C-AD / EPIOS
11	no EIC document specifies a tensor-polarization scale uncertainty, for any species; the 1% requirement is vector-only	ours to specify: $\delta P_{zz}/P_{zz} \leq 5\%$ (Report 2 §6), a 1:1 normalisation on Δ	ANL source + BNL polarimetry
12	relative luminosity between fills of different P_{zz} is unspecified; the $R < 10^{-4}$ requirement is bunch-by-bunch within a fill	ours to specify: $\leq 10^{-3}$ on the two-state ratio of the Δ measurement, together with the detector's cos 2φ′ acceptance harmonic stable to 10^{-4} between the fills — a 10^{-3} difference between the states fakes an amplitude of 5.6×10^{-4}, 2–6 one-year errors (Report 2 §6) — or bunch-by-bunch alternation of the tensor states, which also makes the acceptance cancellation exact; the three-fill b_1 measurement of Report 0 needs 10^{-4}, or its luminosity-corrected estimator	EIC project / ePIC luminosity / source
13	EICrecon has no light-ion far-forward configuration: the transfer matrix is selected by matching the beam proton PDG code	this repository does its own transport; the deliverable is a beamline configuration and a transfer matrix EICrecon can adopt	this programme
14	Roman-Pot commissioning is deferred past year 1 and conditional on beam conditions	projections assume the pots exist from the start	ePIC
15	a 9 GeV electron beam is the early-science baseline; no ESR lattice, emittance, β* or polarization table exists for it	only 5, 10 and 18 GeV are modelled	C-AD
16	the run plan: how a year of running divides between isotopes, ion fills and far-forward optics — no EIC document allocates it, and the Yellow Report luminosity is quoted per configuration, not per measurement	ours to propose (plans/07 WP2): each projection assumes the full luminosity of the year in its own configuration, and a share *f* of that year multiplies every statistical error by $1/\sqrt{f}$, leaves a luminosity quoted as a reach where it is, and multiplies the years to that reach by $1/f$	this programme / EIC project

Table 9 — the assumptions, ranked by leverage on a published number. Items 1–10 and 14–15 are machine and detector inputs, to be asked for precisely; 11, 12 and 16 are properties of a measurement nobody has designed yet and are this programme's to specify; 13 is the point of the development rather than a gap in the specification.

8 · Summary

The electron and proton beams are specified in detail and this programme uses them as specified; the gold rows calibrate the one derivation that matters, the lithium divergence; the ion energies follow from the revolution-period constraint and are checked against the published gold and ^{3}He menus. For lithium itself essentially nothing exists beyond a top energy and a G factor, so §4 is a derivation and not a citation, and its consequence is the sharpest statement this report makes: at every published optics neither the intact ^{6}Li recoil nor the ^{6}Li α spectator can be tagged by its angle (the recoil sits inside whichever of the 10σ envelope and the measured pot aperture binds, the silicon at 5 × 41 and the machine at the other two, §6.1; of the α only the 1.7–2.6% its off-rigidity window and the over-rigid inner branch catch survives, Table 6), and the far-forward half of the ^{6}Li programme exists only with a tagging optics — a one-plane β* de-squeeze of 46–164 with pots that follow it, at 1/6.8–1/12.8 of the luminosity — while the far-forward half of the ^{7}Li programme, the α + t double tag, is already there at the published optics, caught by rigidity rather than by angle, and would lose a net factor 7.7–14.5 under a de-squeeze optimised on its own beam (§5); what goes to C-AD is therefore two run configurations, the tagging optics for

^{6}Li and the published one for ^{7}Li, and one new optics, not two. The detector is specified where it matters most for the tags, in a geometry file that can be read, and least where the central reconstruction lives, which is why Table 8 marks the track angular resolution a placeholder and the electron-identification curve a construction between two non-ePIC sources, rather than giving numbers that would look like a specification.

References

[1] R. Abdul Khalek et al., *Science requirements and detector concepts for the Electron-Ion Collider: EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419 (`refs/2103.05419_part1.pdf` … `refs/2103.05419_part4.pdf`): Tables 10.1 and 10.2 (beam parameters per configuration and optics), Table 11.48 (Roman-Pot p_T smearing), §11.6 (far forward), §14.5.3 (superconducting nanowires), and the detector requirement tables.

[2] G. Atoian et al. (EPIOS), *Realizing the scientific program with polarized ion beams at EIC*, Phys. Rev. C 113 (2026) 060501, arXiv:2510.10794 (`refs/2510.10794.pdf`): Table I (HERA versus EIC parameters); the species table with G and mc^2/G; the synchronisation mechanism (pp. 12–13); the polarized-lithium case.

[3] E. Hamwi, G. H. Hoffstaetter, *Polarization transmission in the Electron-Ion Collider's Hadron Storage Ring*, Phys. Rev. Accel. Beams 29 (2026) 073501, arXiv:2509.18558 (`refs/2509.18558.pdf`): Table I, the species-dependent resonance spectrum; small-G ions cannot use Siberian snakes.

[4] eic/epic, branch main: `compact/far_forward/roman_pots_eRD24_design.xml` , `offM_tracker.xml` and `compact/fields/beamline_*.xml` , read directly 2026-08-26/27: station positions, module size, per-energy insertions, the RF-shield comment.

[5] A. Jentsch (ePIC), *Far-Forward Detectors and Physics with ePIC @ the EIC*, DIS 2023 (`refs/ePIC_far_forward_talk_DIS_2023_v2.pdf`), and the ePIC Collaboration Meeting, JLab, July 2025: acceptances, the AC-LGAD specification, the in-vacuum cooling load, the x-moving layer.

[6] A. Jentsch, Z. Tu, C. Weiss, Phys. Rev. C 104 (2021) 065205, arXiv:2108.08314 (`refs/2108.08314.pdf`): Table I, the far-forward acceptance windows the router uses.

[7] ePIC Collaboration, *Preliminary Design Report*, September 2024 — the 1 October 2024 snapshot of chapters 2 and 8, doi:10.5281/zenodo.13866213; the same material stands in the Preliminary Technical Design Report v3.1 of January 2026, doi:10.5281/zenodo.18271601 — read 2026-09-06: p. 203, Roman-Pot rates, a few Hz per channel in normal running and 30–50 kHz at ≈ 10σ from beam halo; Table 8.1, p. 30, the ePIC tracking requirement, which carries the Yellow Report Fig. 8.3 constants; Figure 8.9, p. 36, the full-simulation $\delta p/p$ ("ePIC 24.08", single pions) in five η ranges.

[8] This repository: Report 1 §6.1 and `evgen/scripts/tagging_optics.py` (the tagging optics priced), Report 2 (the reconstruction chain), Report 4 (the near-beam study), `plans/10_beam_divergence_light_ions.md` (the derivation record), `fastsim/scripts/tagging_acceptance.py` (Table 6), `plans/05_doubly_polarized_generator.md` §5.2 (the cluster momentum density and its β band), `plans/03_phase2_full_simulation.md` §2.2 and `tools/fullsim/README.md` (the far-forward routing and the particle-gun scan).

[9] Electron-Ion Collider, *EIC Requirements*, Hadron Storage Ring (`eic.jlab.org/Requirements/HSR.html`), read 2026-09-02: G-HSR.05 and G-HSR.06, the proton and ^{3}He energy requirements ("41 GeV, and from 100 to 275 GeV"; "41 GeV/nucleon, and from 100 to 183 GeV/nucleon").

[10] S. Maple (ePIC), *Tracking and Inclusive DIS reconstruction with the ePIC detector at the EIC*, seminar, University of Birmingham, 11 December 2024 (`refs/EIC_Seminar_SMaple_2024.pdf`), slide 47: the fully simulated pseudodata fit $\Delta p/p$ [%] = 0.055 p ⊕ 0.45 for the scattered electron from tracking, 0 ≤ η ≤ 0.5, Pythia 8 NC DIS, Craterlake 23.12.0, $Q^2 > 100$ GeV².

[11] F. Willeke et al. (J. Beebe-Wang, ed.), *EIC Conceptual Design Report*, BNL-221006-2021-FORE (2021), doi:10.2172/1765663, read 2026-09-06: the hadron beam-beam design limit — §1.2 p. 5 ("hadrons: $\xi_p \le 0.015$; electrons: $\xi_e \le 0.1$"), §1.4 p. 11 ("suggested by previous performance and confirmed by extensive beam-beam simulations"), §3.1.1 p. 95 ("RHIC has achieved a beam-beam parameter of $\xi_p = 0.015$ in proton-proton collisions; we therefore base the EIC design on the same value") and §4.6.1 pp. 392 and 395; Tables 3.3 and 3.4, whose proton ξ (h/v, 10^{-3}) is 3/3, 12/12, 12/12, 14/14 and 15/9 at 18 × 275, 10 × 275, 10 × 100, 5 × 100 and 5 × 41 GeV. For a lithium beam the CDR does not restate the number, but it does not leave it open either: §1.4 p. 14 scales it ("The beam-beam tune shift for the hadrons is also proportional to Z but is inversely proportional to A"), Table 3.5 runs the Au beam at 1–5 × 10^{-3}, and Appendix A §A.2.2 p. A-12 requires that "the total combined beam-beam tune shift of the two IRs does not exceed 0.03 for ions" — 0.015 per interaction region.

Appendix A · Revision history

DATE	REVISION
2026-09-06	Sourcing pass — the lenses the 2026-09-02 review left unchecked (its §5.3) and its open items 22 and 23. Table 9 row 2's $\xi_p \le 0.015$ is the Conceptual Design Report's hadron beam-beam design limit, now [11] (plans/04 #22 closed). Table 8's tracking row replaces "ePIC full simulation reaches 0.45–0.6% at 1 GeV/c, so realistic to ≈ ±20%" — the 0.6% was the 1.4 T reference design of 2023, not ePIC — by the η-resolved full simulation of the Preliminary Design Report, Figure 8.9: 0.38 / 1.08 / 3.0% at 1 GeV/c in the barrel and the two backward slices against the model's 0.50 / 1.00 / 3.00, so the barrel is ≈ 30% pessimistic and the slices that hold eleven of the twelve sweet spots are right to ≈ 8% ([7] gains its public locator, Maple's 0.45% is [10]; plans/04 #23 closed). Table 8's EMCal-backward and tracking

	source cells name the Yellow Report tables that actually differ backward — Table 11.2 against Fig. 8.3 / Table 8.20 — Fig. 8.3 having been read as an image for the first time.
2026-09-02	Consistency review of the series (`docs/consistency_review_2026-09-02.md`), 37 findings on this report. Sourcing and status: the ion energy menu is quoted from the EIC hadron-ring requirements G-HSR.05 and G-HSR.06, now [9], and GeV/u is defined where it first appears (C005, F234); Table 1's deuteron cap is a design value against the 135 of YR Table 10.3 (C002); Table 3 is the Yellow Report's strong-hadron-cooling table, so the divergences of §4.1 and Table 5 are start-of-store values (C001), its 275 GeV rows are labelled by electron energy, Table 2's EPIOS rows are the 10 × 275 GeV, 1160-bunch column, and the gold calibration and Figure 2's 275 GeV pair rest on the 290-bunch row (F221, F224, F219, F188); the status vocabulary of §1 admits assumption, placeholder, stand-in and proposal, and the Roman-Pot row of Table 7 carries one (F092). §4.2: the current geometry holds the horizontal inner edge at 48, 32 and 16 mm, which puts the approach factor at 6.9, 7.9 and 4.5, the 2.4 cm being the array's own half-height (F001); the companion β* asks are ≈ 150 and ≈ 230 m, and 42 m is a gentler de-squeeze than the LHC has demonstrated (F100, F220); parallel-to-point transport is an assumption at the published optics as well (F135); B = 50 GeV^{-2} carries its 40–60 band and the over-rigid branch the 144 mm module edge (F175, F223); the species factor is 1.52 for ^{7}Li (F225). §5 and Table 6: the tagging-optics rows are re-run 2026-08-29 (F109); the ^{7}Li tagged-sampler figures read 0.9617 and 2.5–2.8%, decomposed to the 0.7 point that separates the two tag definitions (F102, F108, F217, F218); both fragments are recorded in 17–25% of breakups and the ^{7}Li net loss is 7.7–14.5 (F014, F216); the angular scan is four ladders of 117 points (F098). Table 8's twelve sweet spots run η = −3.4 to −0.9 (F134). Table 9: row 5 says the charge sets the margin of the far-forward Z-identification rather than its existence (F222); row 10 gains the ring placeholders $P_z = 0.7$ and $P_{zz} = 0.6$, matched by a new Table 5 row, and the spin direction delivered at IP6 — the vertical axis of the arcs assumed to reach it, a polar tilt costing $\sin^2\theta_S$ and no transverse axis leaving no observable (F034, F137); row 12 gains the cos 2φ′ acceptance harmonic stable to 10^{-4} that the two-state luminosity ratio needs (F008, F130). §8 states that neither ^{6}Li tag is angular, that the ^{7}Li α + t tag needs no de-squeeze, and that what is asked of C-AD is two run configurations (F101, F136). Figures 1 and 2 regenerated: the band label carries three significant figures, the panel letters are left to the captions, and the divergence panel is titled for the coherent acceptance it sets (F228, F231, F233). Two items are left to the author: what sources Table 9 row 2's $\xi_p \le 0.015$ (F226), and what produces the upper end of Table 8's 0.45–0.6% ePIC full-simulation momentum resolution (F237). Figure 1's caption counts the five species plotted (F229), and Table 6's caption separates the retired divergence from the retired circle (F235). Wording suggestions of the same review applied: F037, F040, F042, F049, F076, F077, F091, F099, F103, F107 (the precise 40.8 / 99.5–137.5 beside the rounded menu; the luminosity of every projection named as the flat per-year placeholder rather than Table 3's per-configuration values; the 72.7 μrad line dated at both ends, the boundary it bounds and the day it left the simulation; the triton row hedged to the 0.745 / 0.915 / 0.933 the over-rigid branch carries; the ^{7}Li 1/7.9, 1/14.8 and 1/10.1 labelled as that beam's own optimum against the 1/6.8, 1/12.8 and 1/9.5 it pays in the ^{6}Li optics, in §5 and in Table 5; the pot edges shown as band threshold over measured lever with the ladder closing on them to 1–4%, the 5 × 41 vertical 6.49 mrad marked an extrapolation of a lever rather than an observed edge, and the measured outer edge 2.85 / 3.85 / 4.00 mrad added beside the 5 mrad convention; Table 8's caption saying which side of the chain runs the η table; and the envelope said to bind horizontally. Table 9's run-plan row is restated in the wording Report 2 Table 2 uses for the same share, and the EPIOS column is pointed at Table 2.)
2026-08-29	Verification pass over the round. §5's first conclusion and Table 6's ^{7}Li α entry carry the 5 × 40.8 caveat the caption already stated: the 97% is an angle-only routing, and the measured per-band insertion leaves 57%, station 2 only. Table 6's caption decomposes the ^{6}Li α tag per configuration — 1.51 / 1.51 / 1.54 points in the Roman-Pot window, 0.13 / 0.16 / 1.00 over rigid, 0.20 / 0.01 / 0.08 in the near-beam tail and 0.02 into the B0 at 5 × 40.8 — which sums to the column rather than to a single 1.5%. §6.1's lever ratio is quoted per configuration, 4.2, 6.3 and 10.2, in place of a factor 9; the Z/A = 0.5 lattice edge as the applied 1.61 mrad; the ^{7}Li error-bar penalty as 2.78, 3.81 and 3.15, on the per-configuration tagging levers of that day. The 2026-08-28 row's "1.6% and 1.7%" is labelled as the pre-over-rigid reading. Later the same day the per-configuration transport became the baseline for the tagging optics as well as for the coherent channel, and Table 6's three tagging rows were re-run on it: the ^{6}Li α tag is 0.31 / 0.22 / 0.29 against 0.35 / 0.28 / 0.29, the ^{6}Li d tag 0.54 / 0.46 / 0.57 against 0.57 / 0.51 / 0.57 and the ^{7}Li α tag 0.987 / 0.991 / 0.994 against 0.988 / 0.992 / 0.994, with the abstract, §5 and the derived ranges (22–31% and 46–57%) following. §4.2 now states what the optics asks of the pots in millimetres — an inner edge at 6.99, 4.08 and 3.52 mm against the 48, 32 and 16 mm horizontal edges of the measured silicon — rather than leaving the pot half of the lever qualitative. The 5 × 41 vertical lever R_{34} = 4.56 m, measured through a zeroed-insertion scratch geometry, enters §6.1 and Table 9; and the choice of 5 × 41 lattice is stated as a choice — the ePIC baseline file at twice the field, an educated guess, with the Z/A = 0.5 file as the labelled alternative — so Table 9 row 6 asks the far-forward working group to confirm it rather than to settle a fork the numbers wait on. Money plot 4's b_1-free tagged kernel and per-configuration blind block of the same round move the S + D sampler's Roman-Pot tag from 0.0285 / 0.0250 / 0.0267 to 0.0284 / 0.0247 / 0.0265, so Table 6's two-sampler note now reads 1.5% against 2.5–2.8%.
2026-08-28	Table 9 gained item 16, the run plan: no EIC document allocates a year of running between isotopes, ion fills and far-forward optics, and the Yellow Report luminosity is quoted per configuration rather than per measurement, so the allocation is this programme's to propose (plans/07 WP2). Every projection in this series is quoted at the full year in its own configuration; under a share *f* of that year every statistical error grows as $1/\sqrt{f}$, a luminosity quoted as a reach does not move, and the years needed to accumulate it grow as $1/f$.

2026-08-28	Rewritten as a paper. The far-forward optics are now per configuration throughout the code (`farforward.yr_optics`, `tagging_optics`, a rectangular envelope applied to each fragment's azimuth in the spectator routing — at every aspect ratio since a 2026-08-28 review, an isotropic envelope having until then fallen back on its inscribed circle), and §5 with Table 6 is new: the lithium tags at the Yellow Report optics and at the tagging optics of Report 1 §6.1, replacing the single proton-derived 73 μrad the fast simulation had applied at every configuration, which gave the ^{6}Li α tag as 1.85% at 18 × 137.5 and 20% at 5 × 40.8 against the 1.6% and 1.7% this section read before the over-rigid branch entered every column later the same day; the current Table 6 reads 2.6% and 1.9%. §4.2 states the tagging optics as a proposal; §7 adds the two questions it raises for C-AD and the calorimeter noise floor. β = 0.30 GeV is defined where it is first used, and Table 6's triton row is now the measured over-rigid branch. §5 gained the ^{7}Li luminosity-cost paragraph — the tagging optics is a factor 8–15 net loss for ^{7}Li, so the two isotopes are separate runs — and Table 6's caption the two-sampler note: S wave here, S + D in Report 4 §2.1, 1.5% against 2.5–2.9% (plans/09 B2, B3). The ^{7}Li residual between the two samplers is a difference of acceptance definition and not of density, and like for like they agree to 0.1 point: the pure model's Roman-Pot tag is 0.9568 / 0.9668 / 0.9721, and inside the k ≤ 1.2 GeV/c on which the tagged model's momentum grid ends 0.9626 / 0.9683 / 0.9736, against the tagged sampler's 0.9620 / 0.9678 / 0.9728; at 10 × 99.5 and 18 × 117.9 the B0 fraction is zero and the two definitions differ only by the over-rigid inner branch, 0.2 and 0.3 points. With P_D = 0 the two ^{6}Li samplers agree quantile by quantile. Late the same day the pot aperture was re-measured in the current ePIC geometry (`epic-main` at git 9aaa2969), and §6.1, §7 and every row of Table 6 follow it: the horizontal edge is 2.50 / 1.51 / 0.53 mrad against 2.00 / 1.35 / 1.03, the levers (R_{12}, R_{34}, D) are measured per configuration for the first time, the coherent tag at the Yellow Report optics moves to 9.4×10^{-10} / 6.2×10^{-27} / 7.1×10^{-14}, and the over-rigid triton — 60 of 60 events on Roman-Pot silicon at dx = +66 mm with a ZDC deposit behind it — lifts the ^{7}Li t tag from 0.033 / 0.004 / 0.005 to 0.778 / 0.919 / 0.939, retires "no coverage" and opens the ^{7}Li α + t double tag. The ^{6}Li α tag rises to 0.019 / 0.017 / 0.026 and the d tag to 0.093 / 0.081 / 0.139 through the same branch, applied against the per-configuration 48 / 32 / 16 mm blind block (plans/09 B1, `tools/fullsim/README.md`).
2026-08-27	First version, after the γ-matching correction (plans/10): the ion energy menu, the beam tables, the lithium divergence derived, the far-forward geometry read from the current main branch, the central-detector table fact-checked against the Yellow Report (the tracking resolution is the YR requirement and not a placeholder; the calorimeter forward row is mis-sourced; the noise term is missing), and the list of unknowns with owners. Later the same day: the synchronisation mechanism verbatim from EPIOS and the conflict with Table 10.1's 100 GeV row; store cooling not in the baseline.

Status vocabulary: **design value** — appears in a design document as a parameter; **requirement** — a specification the subsystem must meet; **measured** — test beam, operations, or a simulation of the geometry; **derived** — computed here from published inputs; **unknown** — not published anywhere we could reach. Where no published value settles a parameter, the status names what this programme puts in its place instead — **assumption**, **placeholder** or **stand-in**, or this programme's **proposal** — and Table 8, a fact-check, uses the same column to say how each value fails its source. Beam divergences and Δp/p are Yellow Report Tables 10.1 and 10.2 unless stated; ion energies are computed by `fastsim/polli_fastsim/beams.py` and pinned by `fastsim/tests/test_beams.py`; the far-forward optics by `polli_fastsim/farforward.py` and `fastsim/tests/test_optics_20260828.py`. Built by `reports/build_report.py`.